

GARP Risk & AI — 150 Memory & Theory Questions

Formulas, Names, Definitions, Theorems

Unofficial

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Question

In the K-means algorithm, Euclidean distance between two points P and Q in m dimensions is expressed as:

- A. $\sum_{j=1}^m |x_{jQ} - x_{jP}|$
- B. $\sqrt{\sum_{j=1}^m (x_{jQ} - x_{jP})^2}$
- C. $\max_j |x_{jQ} - x_{jP}|$
- D. $\sum_{j=1}^m (x_{jQ} - x_{jP})$

Answer: B

Explanation

Euclidean distance is the square root of the sum of squared differences = L2-norm. Option A is Manhattan distance (L1). Option C is Chebyshev distance. Option D is signed sum (not a metric).

Question

The Manhattan distance between two points P and Q in m dimensions is also known as:

- A. The L2-norm
- B. The L1-norm
- C. The L_∞ -norm
- D. The cosine norm

Answer: B

Explanation

Manhattan distance = L1-norm = sum of absolute differences. L2-norm is Euclidean; L_∞ -norm is Chebyshev; cosine norm is a similarity metric.

Question

The silhouette score for a data point i is defined as:

- A. $s_i = a_i - b_i$
- B. $s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$
- C. $s_i = \frac{a_i + b_i}{2}$
- D. $s_i = \frac{a_i \cdot b_i}{a_i + b_i}$

Answer: B

Explanation

Silhouette score = $(b_i - a_i) / \max(a_i, b_i)$ where a_i is mean intra-cluster distance and b_i is mean distance to nearest other cluster. Range is $[-1, 1]$.

Question

The silhouette score s_i for any data point lies within which range?

- A. $[0, 1]$
- B. $[-1, 1]$
- C. $[-1, 0]$
- D. $[0, \infty)$

Answer: B

Explanation

Silhouette scores are bounded in $[-1, 1]$. $s \approx 1$ = well-clustered; $s \approx 0$ = boundary; $s < 0$ = possibly mis-clustered.

Question

The Within-Cluster Sum of Squares (WCSS) for cluster k is defined as:

- A. $\sum_{j=1}^{n_k} d_j$
- B. $\sum_{j=1}^{n_k} d_j^2$
- C. $\sqrt{\sum_{j=1}^{n_k} d_j}$
- D. $\max_j d_j$

Answer: B

Explanation

$WCSS_k = \sum_{j=1}^{n_k} d_j^2$ where d_j is the distance from point j to the cluster centroid. Squared distances ensure positive and negative deviations do not cancel.

Question

The total WCSS for all K clusters is given by:

- A. $\sum_{k=1}^K \text{WCSS}_k$
- B. $\prod_{k=1}^K \text{WCSS}_k$
- C. $\max_k \text{WCSS}_k$
- D. $\frac{1}{K} \sum_{k=1}^K \text{WCSS}_k$

Answer: A

Explanation

Total WCSS = sum of individual cluster WCSS values. This is the objective function minimised by K-means.

Question

The expected total prediction error of a model can be decomposed as:

- A. Bias + Variance
- B. $\text{Bias}^2 + \text{Variance}^2$
- C. $\text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$
- D. Bias + Variance + Irreducible Error

Answer: C

Explanation

Total error = $\text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$. The irreducible component represents noise that no model can eliminate.

Question

The Ridge regression loss function is:

- A. $\frac{1}{N} \sum (y_i - \hat{y}_i)^2 + \lambda \sum |\beta_j|$
- B. $\frac{1}{N} \sum (y_i - \hat{y}_i)^2 + \lambda \sum \beta_j^2$
- C. $\sum (y_i - \hat{y}_i)^2$
- D. $\lambda \sum |\beta_j|$

Answer: B

Explanation

Ridge uses L2 penalty: $\lambda \sum \beta_j^2$. LASSO uses L1 penalty: $\lambda \sum |\beta_j|$. Options A and C are LASSO and OLS respectively.

Question

The LASSO regression loss function is:

- A. $\frac{1}{N} \sum (y_i - \hat{y}_i)^2 + \lambda \sum \beta_j^2$
- B. $\frac{1}{N} \sum (y_i - \hat{y}_i)^2 + \lambda \sum |\beta_j|$
- C. $\sum \beta_j^2$
- D. $\sum (y_i - \hat{y}_i)^2$

Answer: B

Explanation

LASSO uses L1 penalty: $\lambda \sum |\beta_j|$. It can drive coefficients to exactly zero, performing feature selection.

Question

The Elastic Net regression loss function combines:

- A. L1 and L2 penalties
- B. Only L1 penalty
- C. Only L2 penalty
- D. Neither penalty

Answer: A

Explanation

Elastic Net = L1 + L2 = $\lambda_1 \sum \beta_j^2 + \lambda_2 \sum |\beta_j|$. It inherits sparsity from L1 and grouping from L2.

Question

Precision is defined as:

- A. $\frac{TP}{TP + FN}$
- B. $\frac{TP}{TP + FP}$
- C. $\frac{TN}{TN + FP}$
- D. $\frac{TP + TN}{TP + TN + FP + FN}$

Answer: B

Explanation

Precision = $TP / (TP + FP)$. Option A is recall (sensitivity); Option C is specificity; Option D is accuracy.

Question

Recall (sensitivity) is defined as:

- A. $\frac{TP}{TP + FP}$
- B. $\frac{TP}{TP + FN}$
- C. $\frac{TN}{TN + FP}$
- D. $\frac{FP}{FP + TN}$

Answer: B

Explanation

Recall = $TP / (TP + FN)$ = true positive rate. Option A = precision; Option C = specificity; Option D = false positive rate.

Question

The F1 score is defined as:

- A. $\frac{P + R}{2}$
- B. $\frac{2 \cdot P \cdot R}{P + R}$
- C. $\sqrt{P \cdot R}$
- D. $P \cdot R$

Answer: B

Explanation

$F1 = \text{harmonic mean of precision and recall} = 2PR/(P + R)$. Option A is arithmetic mean; Option C is geometric mean; Option D is product.

Question

Specificity is defined as:

- A. $\frac{TP}{TP + FN}$
- B. $\frac{TN}{TN + FP}$
- C. $\frac{TP}{TP + FP}$
- D. $\frac{FP}{FP + TN}$

Answer: B

Explanation

Specificity = $TN / (TN + FP)$ = true negative rate. Options A, C, D are recall, precision, and FPR respectively.

Question

Accuracy is defined as:

- A. $\frac{TP + TN}{TP + TN + FP + FN}$
- B. $\frac{TP}{TP + FP}$
- C. $\frac{TP + FN}{FP + FN}$
- D. $\frac{TP + TN}{TP + TN}$

Answer: A

Explanation

Accuracy = (TP+TN)/Total. Option D is the error rate.

Question

Error rate (1 - accuracy) can be expressed as:

- A. $\frac{TP + TN}{Total}$
- B. $\frac{FP + FN}{Total}$
- C. $\frac{TP + FP}{Total}$
- D. $\frac{FN + TN}{Total}$

Answer: B

Explanation

Error rate = $(FP + FN) / Total = 1 - \text{accuracy}$. Options A, C, D are not error rates.

Question

Mean Squared Error is:

- A. $\frac{1}{N} \sum (y_i - \hat{y}_i)$
- B. $\frac{1}{N} \sum |y_i - \hat{y}_i|$
- C. $\frac{1}{N} \sum (y_i - \hat{y}_i)^2$
- D. $\sqrt{\frac{1}{N} \sum (y_i - \hat{y}_i)^2}$

Answer: C

Explanation

MSE = average squared error. Option A would sum to zero; Option B is MAE; Option D is RMSE.

Question

Mean Absolute Error is:

- A. $\frac{1}{N} \sum (y_i - \hat{y}_i)^2$
- B. $\frac{1}{N} \sum |y_i - \hat{y}_i|$
- C. $\sqrt{\frac{1}{N} \sum (y_i - \hat{y}_i)^2}$
- D. $\frac{1}{N} \sum (y_i - \hat{y}_i)$

Answer: B

Explanation

MAE = average absolute error. It is less sensitive to outliers than MSE.

Question

Root Mean Square Error is:

- A. $\frac{1}{N} \sum (y_i - \hat{y}_i)^2$
- B. $\frac{1}{N} \sum |y_i - \hat{y}_i|$
- C. $\sqrt{\frac{1}{N} \sum (y_i - \hat{y}_i)^2}$
- D. $\frac{1}{N} \sum (y_i - \hat{y}_i)$

Answer: C

Explanation

RMSE = square root of MSE. It is expressed in the same units as the target variable.

Question

Mean Absolute Percentage Error is:

- A. $\frac{1}{N} \sum |y_i - \hat{y}_i|$
- B. $100 \cdot \frac{1}{N} \sum \left| \frac{y_i - \hat{y}_i}{y_i} \right|$
- C. $\frac{1}{N} \sum (y_i - \hat{y}_i)^2$
- D. $\sqrt{\frac{1}{N} \sum (y_i - \hat{y}_i)^2}$

Answer: B

Explanation

$\text{MAPE} = 100 \times \text{mean of } |(y_i - \hat{y}_i)/y_i|$. It is scale-independent but fails when actuals are near zero.

Question

The simple linear regression model is:

- A. $y_i = \beta_0 \cdot \beta_1 x_i + u_i$
- B. $y_i = \beta_0 + \beta_1 x_i + u_i$
- C. $y_i = \beta_0^{\beta_1} + u_i$
- D. $y_i = \beta_0 / (\beta_1 x_i) + u_i$

Answer: B

Explanation

Simple linear regression: $y_i = \beta_0 + \beta_1 x_i + u_i$. β_0 is the intercept, β_1 is the slope, u_i is the error term.

Question

The logistic (sigmoid) function is defined as:

- A. $f(z) = z$
- B. $f(z) = e^z$
- C. $f(z) = \frac{1}{1 + e^{-z}}$
- D. $f(z) = \ln(z)$

Answer: C

Explanation

Sigmoid = $1/(1 + e^{-z})$. Output is bounded in $(0, 1)$, making it suitable for probability estimation in logistic regression.

Question

In a logistic regression with m features, the probability that $y_i = 1$ is:

- A. $P_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_m x_{mi}$
- B. $P_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_{1i} + \dots + \beta_m x_{mi})}}$
- C. $P_i = e^{\beta_0 + \beta_1 x_{1i} + \dots + \beta_m x_{mi}}$
- D. $P_i = \ln(\beta_0 + \beta_1 x_{1i} + \dots + \beta_m x_{mi})$

Answer: B

Explanation

Logistic regression applies the sigmoid function to the linear predictor: $P_i = 1/(1 + e^{-(\beta_0 + \beta X)})$.

Question

The log-likelihood function for the logit model is:

- A. $\sum_i [y_i \ln F(y_i) + (1 - y_i) \ln(1 - F(y_i))]$
- B. $\sum_i y_i$
- C. $\sum_i (y_i - F(y_i))^2$
- D. $\prod_i F(y_i)$

Answer: A

Explanation

Log-likelihood sums log-probabilities of observed outcomes. Multiplying probabilities turns into summing logs. Option D is the likelihood (before log); Option C is a squared-error form (not likelihood).

Question

The OLS estimate of the slope in simple linear regression is:

- A. $\hat{\beta}_1 = \bar{y} - \hat{\beta}_0 \bar{x}$
- B. $\hat{\beta}_1 = \frac{\sum y_i x_i - N \bar{y} \bar{x}}{\sum x_i^2 - N \bar{x}^2}$
- C. $\hat{\beta}_1 = \frac{\bar{y}}{\bar{x}}$
- D. $\hat{\beta}_1 = \sum (y_i - \bar{y})(x_i - \bar{x})$

Answer: B

Explanation

OLS slope = covariance of x and y divided by variance of x . Option A is the intercept formula (rearranged). Option C is a naive ratio; Option D is missing the denominator.

Question

Given the OLS slope estimate $\hat{\beta}_1$, the intercept is computed as:

- A. $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
- B. $\hat{\beta}_0 = \bar{y} + \hat{\beta}_1 \bar{x}$
- C. $\hat{\beta}_0 = \bar{y} / \hat{\beta}_1 \bar{x}$
- D. $\hat{\beta}_0 = \hat{\beta}_1 \bar{x}$

Answer: A

Explanation

The regression line passes through the point of means: $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$, so $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$.

Question

The residual sum of squares (RSS) is:

- A. $\sum (y_i - \hat{y}_i)$
- B. $\sum (y_i - \hat{y}_i)^2$
- C. $\sum |y_i - \hat{y}_i|$
- D. $\sqrt{\sum (y_i - \hat{y}_i)^2}$

Answer: B

Explanation

RSS = sum of squared residuals. OLS minimises RSS. Option A would be zero for OLS by construction.

Question

The z-score standardization formula is:

- A. $z = \frac{x - \mu}{\sigma}$
- B. $z = \frac{x - \min}{\max - \min}$
- C. $z = \ln(x)$
- D. $z = x^2$

Answer: A

Explanation

Standardization = $(x - \mu) / \sigma$. Option B is min-max normalization.

Question

The min-max normalization formula is:

- A. $x' = \frac{x - \mu}{\sigma}$
- B. $x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$
- C. $x' = \ln(x)$
- D. $x' = \frac{1}{x}$

Answer: B

Explanation

Min-max scales values to [0, 1]: $x' = (x - x_{\min}) / (x_{\max} - x_{\min})$.

Question

The j -th principal component is expressed as:

- A. $PC_j = b_{1j}P_1 + b_{2j}P_2 + \dots + b_{mj}P_m$
- B. $PC_j = P_1 \cdot P_2 \cdot \dots \cdot P_m$
- C. $PC_j = \max(P_1, P_2, \dots, P_m)$
- D. $PC_j = \frac{P_1 + P_2 + \dots + P_m}{m}$

Answer: A

Explanation

A principal component is a linear combination of standardised predictors weighted by component loadings b_{ij} .

Question

Cosine similarity between two vectors P and Q is:

- A. $\|P - Q\|$
- B. $\frac{P \cdot Q}{\|P\| \|Q\|}$
- C. $\|P\| + \|Q\|$
- D. $\frac{P}{Q}$

Answer: B

Explanation

Cosine similarity = dot product divided by the product of magnitudes. Range is $[-1, 1]$.

Question

The TF-IDF weight of term i in document j is:

- A. $TF_{ij} + IDF_i$
- B. $TF_{ij} \times IDF_i$
- C. TF_{ij} / IDF_i
- D. $TF_{ij} - IDF_i$

Answer: B

Explanation

TF-IDF = term frequency $\times$ inverse document frequency. Term frequency captures within-document frequency; IDF captures rarity across the corpus.

Question

The inverse document frequency of term i across D documents is:

- A. $\ln(D/d_i)$
- B. $D \cdot d_i$
- C. $D + d_i$
- D. $D - d_i$

Answer: A

Explanation

$\text{IDF}_i = \ln(D/d_i)$ where d_i is the number of documents containing term i . If term appears in all documents, $\text{IDF} = 0$.

Question

L2 normalization of a vector v divides each element by:

- A. $\sum v_i$
- B. $\sqrt{\sum v_i^2}$
- C. $\max(v_i)$
- D. $|v_1|$

Answer: B

Explanation

L2 norm = $\sqrt{\sum v_i^2}$. Dividing by it produces a unit vector.

Question

Bayes' rule is written as:

- A. $P(c|d) = P(d|c)P(c) / P(d)$
- B. $P(c|d) = P(c) + P(d)$
- C. $P(c|d) = P(d|c) / P(c)$
- D. $P(c|d) = P(c) \cdot P(d)$

Answer: A

Explanation

Bayes: posterior = likelihood $\times$ prior / evidence. Used in Naive Bayes classification.

Question

The Naive Bayes classification rule (argmax form) is:

- A. $\hat{c} = \arg \max_c P(c)$
- B. $\hat{c} = \arg \max_c P(w_1|c)P(w_2|c) \cdots P(w_N|c)P(c)$
- C. $\hat{c} = \arg \min_c P(c)$
- D. $\hat{c} = \sum_c P(c)$

Answer: B

Explanation

Naive Bayes predicts the class maximising the product of conditional word probabilities and the class prior, exploiting the independence assumption.

Question

The scaled dot-product attention score between query Q_i and key K_j is:

- A. $Q_i \cdot K_j$
- B. $\frac{Q_i \cdot K_j}{\sqrt{d_k}}$
- C. $\frac{Q_i + K_j}{2}$
- D. $|Q_i - K_j|$

Answer: B

Explanation

Scaled dot-product attention divides the dot product by $\sqrt{d_k}$ where d_k is key dimension. This prevents softmax from saturating.

Question

The softmax function is defined as:

- A. $\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$
- B. $\text{softmax}(x_i) = x_i / \sum_j x_j$
- C. $\text{softmax}(x_i) = e^{x_i}$
- D. $\text{softmax}(x_i) = \ln(x_i)$

Answer: A

Explanation

Softmax normalises exponentials into a probability distribution. Used in attention and multiclass outputs.

Question

The gradient descent weight update rule is:

- A. $w_i^{\text{new}} = w_i^{\text{old}} + \eta \frac{\partial L}{\partial w_i}$
- B. $w_i^{\text{new}} = w_i^{\text{old}} - \eta \frac{\partial L}{\partial w_i}$
- C. $w_i^{\text{new}} = w_i^{\text{old}} \cdot \eta$
- D. $w_i^{\text{new}} = \eta$

Answer: B

Explanation

Gradient descent subtracts the learning-rate-scaled gradient. Option A would be gradient ascent.

Question

Exponential decay of the learning rate is:

- A. $\eta_t = \eta_0 + kt$
- B. $\eta_t = \eta_0 e^{-kt}$
- C. $\eta_t = \eta_0 \cdot kt$
- D. $\eta_t = \eta_0 / k$

Answer: B

Explanation

Exponential decay: $\eta_t = \eta_0 e^{-kt}$, where k controls decay and t is the epoch.

Question

Inverse decay of the learning rate is:

- A. $\eta_t = \eta_0 e^{-kt}$
- B. $\eta_t = \eta_0 / (1 + kt)$
- C. $\eta_t = \eta_0 (1 + kt)$
- D. $\eta_t = \eta_0 \cdot k$

Answer: B

Explanation

Inverse decay: $\eta_t = \eta_0 / (1 + kt)$. Alternative to exponential decay for stochastic gradient descent.

Question

The K-means algorithm is also known as:

- A. Lloyd's algorithm
- B. Newton's algorithm
- C. Bayes' algorithm
- D. Fisher's algorithm

Answer: A

Explanation

K-means is also known as Lloyd's algorithm, after Stuart Lloyd who developed it at Bell Labs in 1957.

Question

The fairness impossibility theorem showing that demographic parity, predictive rate parity, and equal opportunity cannot all hold under unequal base rates is attributed to:

- A. Kleinberg, Mullainathan, and Raghavan
- B. Fisher, Neyman, and Pearson
- C. Bentham and Mill
- D. Kant and Aristotle

Answer: A

Explanation

Kleinberg, Mullainathan, and Raghavan (2016) proved this impossibility result. Chouldechova independently derived a related result.

Question

Which pair of philosophers are considered the founders of utilitarianism?

- A. Kant and Aristotle
- B. Bentham and Mill
- C. Rawls and Nozick
- D. Plato and Socrates

Answer: B

Explanation

Jeremy Bentham and John Stuart Mill developed classical utilitarianism in the 18th and 19th centuries.

Question

The concept of the categorical imperative is attributed to:

- A. Immanuel Kant
- B. John Stuart Mill
- C. Jeremy Bentham
- D. Aristotle

Answer: A

Explanation

Kant's categorical imperative: act only on maxims that could be universal laws.

Question

Virtue ethics is most closely associated with which ancient philosopher?

- A. Aristotle
- B. Kant
- C. Mill
- D. Bentham

Answer: A

Explanation

Virtue ethics has roots in ancient Greek philosophy, particularly Aristotle's *Nicomachean Ethics*.

Question

The 2017 paper introducing the transformer architecture is titled:

- A. "Attention Is All You Need"
- B. "Deep Learning for AI"
- C. "Neural Networks and Learning"
- D. "Machine Learning Fundamentals"

Answer: A

Explanation

"Attention Is All You Need" by Vaswani et al. (2017) introduced the transformer architecture based on self-attention.

Question

BERT stands for:

- A. Bidirectional Encoder Representations from Transformers
- B. Bayesian Encoder for Recursive Tasks
- C. Binary Encoder Representation Technique
- D. Bidirectional Encoder for Random Text

Answer: A

Explanation

BERT = Bidirectional Encoder Representations from Transformers. It is an encoder-only model released by Google in 2018.

Question

GPT stands for:

- A. General Purpose Transformer
- B. Generative Pre-trained Transformer
- C. Gradient Pre-training Technique
- D. Global Parameter Training

Answer: B

Explanation

GPT = Generative Pre-trained Transformer. It is a decoder-only autoregressive model family from OpenAI.

Question

BART stands for:

- A. Bidirectional and Auto-Regressive Transformer
- B. Bayesian Auto-Regressive Technique
- C. Binary Auto-Regressive Transformer
- D. Bidirectional Attention for Recognition Tasks

Answer: A

Explanation

BART = Bidirectional and Auto-Regressive Transformer. It combines an encoder and decoder and is used for both understanding and generation tasks.

Question

LSTM stands for:

- A. Long Short-Term Memory
- B. Linear Sequential Time Model
- C. Layered Stochastic Transformation Model
- D. Low-Signal Temporal Method

Answer: A

Explanation

LSTM = Long Short-Term Memory. It is a type of RNN that mitigates the vanishing gradient problem via gating mechanisms.

Question

DBSCAN stands for:

- A. Density-Based Spatial Clustering of Applications with Noise
- B. Data-Based Sample Clustering Algorithm for Networks
- C. Dynamic Bayesian Segmentation Clustering Algorithm
- D. Distributed Batch Spectral Clustering Approach

Answer: A

Explanation

DBSCAN = Density-Based Spatial Clustering of Applications with Noise. It identifies core, border, and noise points using ϵ and minPts.

Question

LASSO stands for:

- A. Least Absolute Shrinkage and Selection Operator
- B. Linear Analysis of Statistical Sparse Outcomes
- C. Layered Adaptive Sparse Subset Optimization
- D. Least Squares Solution Operator

Answer: A

Explanation

LASSO = Least Absolute Shrinkage and Selection Operator. It uses an L1 penalty and can zero out coefficients.

Question

KNN stands for:

- A. K-Nearest Neighbors
- B. Kernel Neural Network
- C. K-Normalized Nodes
- D. Knowledge Notation Network

Answer: A

Explanation

KNN = K-Nearest Neighbors. A non-parametric classifier/regressor based on nearest training examples.

Question

SVM stands for:

- A. Support Vector Machine
- B. Supervised Vector Model
- C. Statistical Variance Method
- D. Sparse Vector Mechanism

Answer: A

Explanation

SVM = Support Vector Machine. It finds the maximum-margin hyperplane separating classes.

Question

CBoW stands for:

- A. Continuous Bag of Words
- B. Complex Bag of Words
- C. Categorical Bag of Words
- D. Continuous Batch of Words

Answer: A

Explanation

CBoW = Continuous Bag of Words. One of two Word2Vec architectures: many inputs, one output.

Question

RAG stands for:

- A. Retrieval Augmented Generation
- B. Random Attention Gate
- C. Recurrent Adversarial Graph
- D. Regularized Attention Gradient

Answer: A

Explanation

RAG = Retrieval Augmented Generation. It augments LLM responses with retrieved documents for factual accuracy.

Question

RLHF stands for:

- A. Reinforcement Learning from Human Feedback
- B. Recursive Layered Hidden Features
- C. Random Loss with High Frequency
- D. Regularized Linear Hierarchical Framework

Answer: A

Explanation

RLHF = Reinforcement Learning from Human Feedback. Used in LLM training to align outputs with human preferences.

Question

PCA stands for:

- A. Principal Component Analysis
- B. Predictive Cluster Aggregation
- C. Pattern Classification Algorithm
- D. Probabilistic Component Approximation

Answer: A

Explanation

PCA = Principal Component Analysis. A dimensionality reduction technique using orthogonal linear combinations.

Question

SR 11-7, "Guidance on Model Risk Management", was issued by which U.S. regulator?

- A. The Federal Reserve and the OCC
- B. The SEC
- C. The FDIC alone
- D. The CFTC

Answer: A

Explanation

SR 11-7 (2011) was issued jointly by the Federal Reserve and the Office of the Comptroller of the Currency (OCC) as supervisory guidance on model risk management.

Question

Which GDPR article requires notification of a personal data breach to the supervisory authority within 72 hours?

- A. Article 17
- B. Article 33
- C. Article 6
- D. Article 22

Answer: B

Explanation

Article 33 = breach notification. Article 17 = right to erasure; Article 6 = lawful basis; Article 22 = automated decision-making rights.

Question

The GDPR right to erasure (right to be forgotten) is contained in:

- A. Article 6
- B. Article 17
- C. Article 33
- D. Article 5

Answer: B

Explanation

Article 17 = right to erasure. Article 5 = principles; Article 6 = lawful basis; Article 33 = breach notification.

Question

The NIST AI Risk Management Framework is organised around four core functions:

- A. Train, Test, Deploy, Monitor
- B. Govern, Map, Measure, Manage
- C. Design, Build, Audit, Retire
- D. Plan, Do, Check, Act

Answer: B

Explanation

NIST AI RMF = Govern (structures), Map (context), Measure (assessment), Manage (mitigation).

Question

The Three Lines of Defence model in financial risk management is best associated with:

- A. The Institute of Internal Auditors (IIA)
- B. The Federal Reserve alone
- C. The Basel Committee alone
- D. The SEC alone

Answer: A

Explanation

The IIA formalised the Three Lines of Defence model, later updated to the Three Lines Model in 2020.

Question

Inscrutability, as used in AI risk, is best defined as:

- A. The inability to understand the model's internal mechanisms
- B. The inability to explain a single prediction
- C. The inability to generalise out-of-sample
- D. The inability to measure accuracy

Answer: A

Explanation

Inscrutability = opaque internal mechanisms. Explainability = specific decisions. Generalisability = out-of-sample performance.

Question

Interpretability refers to:

- A. Providing reasons for a specific decision
- B. Understanding the model's internal logic and mechanisms
- C. Measuring accuracy
- D. Reducing features

Answer: B

Explanation

Interpretability = internal understanding. Explainability = specific-decision justification.

Question

Explainability is best defined as:

- A. Understanding internal model logic
- B. Providing human-understandable reasons for specific decisions
- C. Model training efficiency
- D. Feature reduction

Answer: B

Explanation

Explainability = post-hoc justification of specific decisions in human terms.

Question

Transparency in AI refers to:

- A. Understanding internal model logic
- B. Openness about the model's design, algorithms, data, and decision-making processes
- C. Explaining specific predictions
- D. Reducing model size

Answer: B

Explanation

Transparency = openness about design, algorithms, data, and processes. Distinct from interpretability (internal logic) and explainability (specific decisions).

Question

Opaqueness in AI refers to:

- A. Lack of transparency in a model's design, data, or decision-making
- B. Model generalisability
- C. Model accuracy
- D. Model speed

Answer: A

Explanation

Opaqueness = lack of transparency. Forms include secrecy (unaware of profiling), confidentiality (internals hidden), and specialised knowledge (requires expertise).

Question

Automation bias is best defined as:

- A. A tendency to over-rely on automated systems, reducing vigilance
- B. Systematic measurement error
- C. Random sampling error
- D. Non-linear relationship in data

Answer: A

Explanation

Automation bias = over-reliance on automated recommendations, reducing critical oversight and human verification.

Question

Proxy discrimination occurs when:

- A. A feature correlated with a protected attribute carries discriminatory effects even after the protected attribute is removed
- B. The model is too complex
- C. The dataset is small
- D. The model uses too many features

Answer: A

Explanation

Proxy discrimination = removing a protected attribute but retaining correlated features that perpetuate discrimination (e.g., ZIP codes as race proxies).

Question

Historical bias in AI refers to:

- A. Random noise in the training set
- B. Past discrimination encoded in training data, which the model then perpetuates
- C. Measurement error
- D. Sampling noise

Answer: B

Explanation

Historical bias arises when training data reflects past discriminatory patterns. Models trained on such data reinforce and amplify these patterns.

Question

Sampling bias occurs when:

- A. The sample is not representative of the target population
- B. The model is too simple
- C. The data has outliers
- D. The model overfits

Answer: A

Explanation

Sampling bias = unrepresentative training sample. The model performs poorly when deployed on the actual target population.

Question

Data composition bias arises from:

- A. Imbalanced class distribution in the training data
- B. Features measured differently across groups
- C. Outliers in the data
- D. Model overfitting

Answer: A

Explanation

Data composition bias = imbalance where minority classes are underrepresented. The model underperforms on minority groups.

Question

Measurement bias occurs when:

- A. Features are measured differently across groups
- B. The class distribution is imbalanced
- C. The sample is small
- D. The model is nonlinear

Answer: A

Explanation

Measurement bias = inconsistent measurement methods across groups (e.g., different diagnostic tests for different populations).

Question

Individual fairness is best defined as:

- A. Equal outcomes across demographic groups
- B. Treating similar individuals similarly
- C. Equal true positive rates
- D. Equal precision

Answer: B

Explanation

Individual fairness = treat like cases alike. Rooted in Aristotle's doctrine. Contrasts with group fairness (statistical parity).

Question

Demographic parity requires:

- A. Equal true positive rates across groups
- B. Equal predicted positive rates across groups
- C. Equal precision across groups
- D. Equal accuracy across groups

Answer: B

Explanation

Demographic parity: $P(\hat{Y} = 1|A = 0) = P(\hat{Y} = 1|A = 1)$. It does not consider whether predictions are correct.

Question

Equal opportunity requires:

- A. Equal positive prediction rates
- B. Equal true positive rates across groups
- C. Equal precision across groups
- D. Equal false positive rates

Answer: B

Explanation

Equal opportunity: $P(\hat{Y} = 1 | Y = 1, A = 0) = P(\hat{Y} = 1 | Y = 1, A = 1)$. Requires equal TPR across groups.

Question

Predictive rate parity requires:

- A. Equal true positive rates across groups
- B. Equal precision across groups
- C. Equal positive prediction rates
- D. Equal accuracy

Answer: B

Explanation

Predictive rate parity: $P(Y = 1 | \hat{Y} = 1, A = 0) = P(Y = 1 | \hat{Y} = 1, A = 1)$. Requires equal precision across groups.

Question

The fairness impossibility theorem states that when base rates differ:

- A. All fairness metrics can be satisfied
- B. Demographic parity, predictive rate parity, and equal opportunity cannot all be satisfied simultaneously
- C. Only equal opportunity can be satisfied
- D. Accuracy is the only relevant metric

Answer: B

Explanation

Kleinberg, Mullainathan, and Raghavan (2016): under unequal base rates, these three fairness criteria are mutually incompatible.

Question

Nonmaleficence refers to:

- A. Actively promoting welfare
- B. Avoiding harm and unjustified risk
- C. Fairness
- D. Respecting autonomy

Answer: B

Explanation

Nonmaleficence = do no harm. It forbids imposing unjustified risks.

Question

Beneficence refers to:

- A. Avoiding harm
- B. Actively promoting the welfare of others
- C. Respecting autonomy
- D. Ensuring fairness

Answer: B

Explanation

Beneficence = do good. It requires positive action to help others.

Question

Autonomy as an AI ethics principle requires:

- A. Respecting individual decision-making
- B. Ensuring fairness
- C. Avoiding harm
- D. Benefiting others

Answer: A

Explanation

Autonomy = respect for individual self-determination and consent.

Question

Justice as an AI ethics principle primarily concerns:

- A. Fairness, equality, and impartiality
- B. Avoiding harm
- C. Benefiting others
- D. Respecting autonomy

Answer: A

Explanation

Justice = fairness, equality, impartiality. Distributive justice concerns allocation; procedural justice concerns fair process.

Question

Consequentialism is an ethical framework that judges actions by:

- A. Adherence to rules
- B. Outcomes or consequences
- C. Character traits
- D. Legal compliance

Answer: B

Explanation

Consequentialism judges by outcomes. Utilitarianism is the most common form, maximising overall welfare.

Question

Deontology is an ethical framework that judges actions by:

- A. Outcomes
- B. Adherence to duties and rules
- C. Character
- D. Profit

Answer: B

Explanation

Deontology judges by duties and rules irrespective of consequences. Kant is the primary proponent.

Question

Virtue ethics focuses on:

- A. Outcomes
- B. Rules
- C. Character traits and living a good life
- D. Legal compliance

Answer: C

Explanation

Virtue ethics asks what kind of person one should be and emphasises virtues like wisdom, courage, and justice.

Question

Contextual integrity requires that:

- A. Data can be freely reused
- B. Data use must respect the norms of the context in which it was provided
- C. Data must be encrypted
- D. Data must be deleted

Answer: B

Explanation

Contextual integrity (Nissenbaum): data provided in one context should not be used in an incompatible context.

Question

Data minimisation requires:

- A. Collecting all available data
- B. Collecting only data necessary for the specified purpose
- C. Sharing data freely
- D. Retaining data indefinitely

Answer: B

Explanation

Data minimisation = collect only what is necessary for the specified purpose. Core GDPR principle.

Question

The right to be forgotten allows data subjects to:

- A. Access their data
- B. Request erasure of personal data under certain conditions
- C. Sell their data
- D. Share data without consent

Answer: B

Explanation

Right to erasure (GDPR Art. 17): allows deletion when data is no longer necessary, consent is withdrawn, etc.

Question

Data provenance refers to:

- A. Data storage format
- B. The origin, history, and legal right to use data
- C. Data encryption
- D. Data classification

Answer: B

Explanation

Data provenance = origin and legal rights to use data. Critical for alternative data and AI training sets.

Question

Data classification is the process of:

- A. Sorting data alphabetically
- B. Categorising data by sensitivity and structure to determine handling and controls
- C. Encrypting data
- D. Compressing data

Answer: B

Explanation

Data classification assigns sensitivity labels (public, internal, confidential, restricted) and structure labels.

Question

Metadata is best defined as:

- A. Raw data
- B. Data about data
- C. Encrypted data
- D. Synthetic data

Answer: B

Explanation

Metadata = data about data. It documents origin, structure, definitions, and relationships.

Question

Role-Based Access Control (RBAC) grants access based on:

- A. Individual discretion
- B. User roles within the organisation
- C. Random selection
- D. Data size

Answer: B

Explanation

RBAC = permissions based on organisational roles. ABAC = attribute-based; MAC = mandatory; DAC = discretionary.

Question

Attribute-Based Access Control (ABAC) grants access based on:

- A. User roles
- B. Attributes of user, resource, and environment
- C. Random selection
- D. Seniority

Answer: B

Explanation

ABAC uses attributes (user, resource, environment) to make access decisions. More flexible than RBAC.

Question

A model inventory is:

- A. A list of all models in use, with owners, risk tiers, and validation status
- B. A list of data sources
- C. A list of employees
- D. A list of IT systems

Answer: A

Explanation

Model inventory = registry of all models. Includes owner, purpose, risk tier, validation status, and interdependencies.

Question

A model landscape is:

- A. A list of all models
- B. A map of interactions and interdependencies between models
- C. A training dataset
- D. A model architecture

Answer: B

Explanation

Model landscape = map of interactions and dependencies. It reveals cascade risk and critical nodes.

Question

Model risk appetite is:

- A. The set of models in production
- B. The level of model risk the organisation is willing to accept
- C. The cost of modelling
- D. The number of models

Answer: B

Explanation

Model risk appetite = acceptable level of model risk, set by the board, guiding governance.

Question

The use test in model validation is:

- A. A quantitative accuracy metric
- B. A qualitative assessment of whether the model is used appropriately for its intended purpose
- C. A code review
- D. A speed test

Answer: B

Explanation

The use test assesses whether the model is used correctly and appropriately in practice. It is qualitative and people-focused.

Question

Outcomes analysis in model validation is used to:

- A. Speed up training
- B. Compare model predictions against realised outcomes to detect bias, degradation, or incorrect assumptions
- C. Reduce features
- D. Increase accuracy

Answer: B

Explanation

Outcomes analysis compares model predictions against actual outcomes to validate model quality in production.

Question

K-means++ is an improved initialisation method whose purpose is to:

- A. Reduce K
- B. Spread initial centroids far apart to reduce poor local optima
- C. Increase WCSS
- D. Eliminate outliers

Answer: B

Explanation

K-means++ selects initial centroids with probability proportional to squared distance from nearest existing centroid, spreading them out and reducing poor local optima.

Question

A scree plot displays:

- A. WCSS against number of clusters K
- B. Data against time
- C. Two features against each other
- D. Model accuracy against training time

Answer: A

Explanation

A scree plot charts WCSS (inertia) against K. The elbow point suggests the optimal K.

Question

A dendrogram is:

- A. A scatter plot
- B. A tree-like diagram showing hierarchical clustering steps
- C. A line chart of loss
- D. A histogram

Answer: B

Explanation

A dendrogram visualises hierarchical clustering: x-axis shows data points, y-axis shows merge distance. Long vertical lines indicate significant cluster boundaries.

Question

A standard confusion matrix contains four cells:

- A. TP, TN, FP, FN
- B. A, B, C, D
- C. Alpha, Beta, Gamma, Delta
- D. Mean, Median, Mode, Range

Answer: A

Explanation

Confusion matrix cells: True Positive, True Negative, False Positive (Type I error), False Negative (Type II error).

Question

The ROC curve plots which two quantities on its axes?

- A. Precision vs recall
- B. True positive rate vs false positive rate
- C. Train error vs test error
- D. Loss vs iteration

Answer: B

Explanation

ROC: y-axis = TPR (sensitivity); x-axis = FPR ($1 - \text{specificity}$). AUC summarises discrimination.

Question

The AUC ranges over which values, and what does $AUC = 0.5$ indicate?

- A. $[0, 1]$; $AUC = 0.5$ indicates random classifier
- B. $[-1, 1]$; $AUC = 0.5$ indicates perfect classifier
- C. $[0, 100]$; $AUC = 0.5$ indicates perfect classifier
- D. $[1, 10]$; $AUC = 0.5$ indicates random classifier

Answer: A

Explanation

AUC ranges in $[0, 1]$. $AUC = 0.5$ = random; $AUC = 1.0$ = perfect; $AUC < 0.5$ = worse than random.

Question

In k -fold cross-validation, the data is:

- A. Split into k equal folds; the model is trained k times, each time holding out a different fold for validation
- B. Split into 2 folds only
- C. Not split at all
- D. Split randomly without reuse

Answer: A

Explanation

k -fold CV: partition into k folds; iterate k times, each time training on $k - 1$ folds and validating on the held-out fold. Average the results.

Question

Stratified cross-validation ensures that:

- A. Each fold has the same class proportion as the whole dataset
- B. Only the majority class is used
- C. All folds have equal mean
- D. Random assignment is avoided

Answer: A

Explanation

Stratified CV preserves class ratios in every fold, critical for imbalanced datasets.

Question

Bootstrapping in model evaluation is:

- A. Sampling with replacement to create new datasets
- B. Sampling without replacement
- C. Splitting data into two halves
- D. Training without validation

Answer: A

Explanation

Bootstrapping draws samples with replacement. About 63% unique observations per sample; 37% out-of-bootstrap.

Question

In bootstrapping, the expected fraction of observations NOT sampled in each bootstrap sample is approximately:

- A. 10%
- B. 25%
- C. 37%
- D. 63%

Answer: C

Explanation

The expected out-of-bootstrap fraction is $1 - 1/e \approx 0.368 = 37\%$.

Question

Grid search is a technique for:

- A. Exhaustively searching through a predefined set of hyperparameter combinations
- B. Removing outliers
- C. Feature selection only
- D. Clustering

Answer: A

Explanation

Grid search evaluates every combination of hyperparameters within specified ranges. Often combined with cross-validation.

Question

The learning rate η in gradient descent controls:

- A. The number of iterations
- B. The step size of each weight update
- C. The number of features
- D. The batch size

Answer: B

Explanation

Learning rate η scales the gradient to determine step size. Too small = slow; too large = overshoot.

Question

Backpropagation is:

- A. Forward-only computation
- B. An algorithm that computes gradients layer-by-layer using the chain rule, starting from the output layer
- C. A clustering method
- D. A type of regularisation

Answer: B

Explanation

Backpropagation computes gradients efficiently by applying the chain rule from output to input layers.

Question

An epoch in neural network training is:

- A. One gradient update
- B. One full pass through the entire training dataset
- C. One validation step
- D. One feature selection step

Answer: B

Explanation

An epoch = one full pass through the training data. Typically many epochs are required for convergence.

Question

Batch gradient descent updates weights using:

- A. One random training example per step
- B. A subset of training examples per step
- C. The entire training set per step
- D. No training examples

Answer: C

Explanation

Batch GD uses the full dataset per update. Stochastic uses one example; mini-batch uses a subset.

Question

Stochastic gradient descent (SGD) updates weights using:

- A. The full training set
- B. One random training example per step
- C. A fixed subset
- D. No examples

Answer: B

Explanation

SGD uses one randomly selected example per update. It is noisier but faster per pass.

Question

Mini-batch gradient descent updates weights using:

- A. One example
- B. The full dataset
- C. A small subset of the training data per step
- D. No data

Answer: C

Explanation

Mini-batch GD uses a small subset (e.g., 32 or 64 examples). It balances parallelism and stochasticity.

Question

Momentum in gradient descent primarily:

- A. Reduces the learning rate
- B. Smooths the optimisation path and accelerates in consistent gradient directions
- C. Eliminates local minima
- D. Converts non-convex to convex

Answer: B

Explanation

Momentum accumulates a velocity vector, accelerating consistent directions and dampening oscillation.

Question

The Adam optimiser combines which two ideas?

- A. Momentum and adaptive learning rates
- B. Batch and stochastic descent
- C. Ridge and LASSO
- D. Precision and recall

Answer: A

Explanation

Adam = Adaptive Moment Estimation. It combines momentum (first moment) with adaptive per-parameter learning rates (second moment).

Question

Dropout regularisation works by:

- A. Removing whole layers
- B. Randomly deactivating a fraction of neurons during training to prevent co-adaptation
- C. Reducing the learning rate
- D. Removing features

Answer: B

Explanation

Dropout randomly drops units during training. At inference, all units are used with scaled weights.

Question

Batch normalisation normalises:

- A. Weights
- B. Intermediate layer inputs, stabilising training and mitigating internal covariate shift
- C. Features
- D. Outputs

Answer: B

Explanation

Batch norm normalises the activations of a layer across the batch, improving stability and speed of training.

Question

Early stopping is a regularisation technique that:

- A. Stops training when validation performance starts to deteriorate
- B. Stops training after a fixed number of epochs
- C. Stops training immediately
- D. Never stops training

Answer: A

Explanation

Early stopping monitors validation loss and halts training when it stops improving, preventing overfitting.

Question

The vanishing gradient problem occurs when:

- A. Gradients grow unboundedly
- B. Gradients become extremely small as they propagate through many layers
- C. Weights are initialised poorly
- D. The learning rate is too high

Answer: B

Explanation

Vanishing gradient: gradients shrink exponentially through deep networks with saturating activations (e.g., sigmoid).

Question

The exploding gradient problem occurs when:

- A. Gradients shrink to near zero
- B. Gradients grow extremely large across layers, destabilising training
- C. Weights are zero
- D. Learning rate is zero

Answer: B

Explanation

Exploding gradient: gradients grow exponentially. Remedies include gradient clipping and careful initialisation.

Question

The ReLU (Rectified Linear Unit) activation function is:

- A. $f(x) = \max(0, x)$
- B. $f(x) = 1/(1 + e^{-x})$
- C. $f(x) = \tanh(x)$
- D. $f(x) = x^2$

Answer: A

Explanation

ReLU = $\max(0, x)$. It is computationally efficient and reduces vanishing gradient for positive values.

Question

The sigmoid activation function is:

- A. $f(x) = \max(0, x)$
- B. $f(x) = 1/(1 + e^{-x})$
- C. $f(x) = \tanh(x)$
- D. $f(x) = x$

Answer: B

Explanation

Sigmoid = $1/(1 + e^{-x})$, bounded in $(0, 1)$. Used in binary outputs but can suffer from vanishing gradients.

Question

The tanh (hyperbolic tangent) activation function is:

- A. $f(x) = \max(0, x)$
- B. $f(x) = 1/(1 + e^{-x})$
- C. $f(x) = (e^x - e^{-x})/(e^x + e^{-x})$
- D. $f(x) = x^2$

Answer: C

Explanation

$\tanh = (e^x - e^{-x})/(e^x + e^{-x})$. Output is bounded in $[-1, 1]$ and centred at zero.

Question

Softmax is used at the output layer primarily for:

- A. Binary classification
- B. Multi-class classification, converting logits into a probability distribution
- C. Regression
- D. Clustering

Answer: B

Explanation

Softmax outputs sum to one, making it appropriate for multi-class probability outputs.

Question

A Type I error in a classification context corresponds to:

- A. False positive
- B. False negative
- C. True positive
- D. True negative

Answer: A

Explanation

Type I error = false positive (rejecting a true null hypothesis). Type II error = false negative.

Question

A Type II error in a classification context corresponds to:

- A. False positive
- B. False negative
- C. True positive
- D. True negative

Answer: B

Explanation

Type II error = false negative (failing to reject a false null hypothesis).

Question

Which of the following is NOT a standard technique for handling class imbalance?

- A. Oversampling the minority class
- B. Undersampling the majority class
- C. Using a weighted loss function
- D. Removing the minority class entirely

Answer: D

Explanation

Removing the minority class destroys the signal and defeats the purpose. Options A, B, C are standard approaches.

Question

SMOTE stands for:

- A. Synthetic Minority Over-sampling Technique
- B. Statistical Missing Value Estimation
- C. Sparse Matrix Optimisation Technique
- D. Sequential Model Ensemble

Answer: A

Explanation

SMOTE = Synthetic Minority Over-sampling Technique. It generates synthetic minority samples via interpolation between existing minority points.

Question

Bagging (Bootstrap Aggregation) is:

- A. Training models sequentially
- B. Training multiple models on bootstrap samples and averaging predictions
- C. Training a single model on all data
- D. Removing features

Answer: B

Explanation

Bagging trains models in parallel on bootstrap samples and averages outputs. Reduces variance.

Question

Random forest differs from bagging by:

- A. Using sequential training
- B. Adding random feature subsetting at each split
- C. Using only one tree
- D. Removing randomness

Answer: B

Explanation

Random forest = bagging + random feature subsampling at each split. Reduces correlation across trees.

Question

Boosting trains models:

- A. In parallel on bootstrap samples
- B. Sequentially, each correcting the errors of the previous model
- C. Randomly
- D. Without any error correction

Answer: B

Explanation

Boosting fits models sequentially, each new model focused on prior residuals or misclassified points. Reduces bias.

Question

AdaBoost is a boosting method that:

- A. Fits residuals
- B. Reweights misclassified training examples and combines weak learners
- C. Uses random features
- D. Removes features

Answer: B

Explanation

AdaBoost reweights training examples after each iteration, focusing subsequent weak learners on previously misclassified points.

Question

Gradient boosting fits new trees to:

- A. The original labels
- B. The residuals of previous trees
- C. Random labels
- D. Bootstrap samples only

Answer: B

Explanation

Gradient boosting fits each new tree to the negative gradient (residuals) of the loss function from prior trees.

Question

XGBoost stands for:

- A. Extreme Gradient Boosting
- B. Extended Gradient Boost
- C. Extra Gradient Boost
- D. Exponential Gradient Boost

Answer: A

Explanation

XGBoost = Extreme Gradient Boosting. An efficient implementation of gradient boosting with regularisation.

Question

Which statement correctly distinguishes bagging and boosting in terms of bias and variance?

- A. Bagging mainly reduces variance; boosting mainly reduces bias
- B. Bagging mainly reduces bias; boosting mainly reduces variance
- C. Both reduce bias only
- D. Both reduce variance only

Answer: A

Explanation

Bagging averages decorrelated models, reducing variance. Boosting sequentially corrects errors, reducing bias.

Question

Lowering the classification threshold generally:

- A. Increases recall and decreases precision
- B. Increases precision and decreases recall
- C. Increases both
- D. Decreases both

Answer: A

Explanation

Lower threshold = more positives predicted = higher recall but lower precision (more false positives).

Question

In logistic regression, the log-odds (logit) of the outcome is:

- A. $\ln(P/(1 - P)) = \beta_0 + \beta_1 x_1 + \dots$
- B. $P/(1 - P)$
- C. $P \cdot (1 - P)$
- D. $\ln(P)$

Answer: A

Explanation

Logit = $\ln(P/(1 - P))$, linear in the coefficients. Logistic regression models the log-odds as a linear function of features.

Question

In logistic regression, the exponential of a coefficient $\exp(\beta_j)$ represents:

- A. The probability
- B. The odds ratio for a one-unit increase in x_j
- C. The variance
- D. The residual

Answer: B

Explanation

$\exp(\beta_j)$ = odds ratio. A one-unit increase in x_j multiplies the odds by $\exp(\beta_j)$.

Question

In LDA, the discriminant function for class j is:

- A. $\delta_j(x) = x^T \hat{\Sigma}^{-1} \hat{\mu}_j - \frac{1}{2} \hat{\mu}_j^T \hat{\Sigma}^{-1} \hat{\mu}_j + \ln(\hat{\pi}_j)$
- B. $\delta_j(x) = x$
- C. $\delta_j(x) = x^2$
- D. $\delta_j(x) = 1/(1 + e^{-x})$

Answer: A

Explanation

LDA discriminant: $\delta_j(x) = x^T \hat{\Sigma}^{-1} \hat{\mu}_j - \frac{1}{2} \hat{\mu}_j^T \hat{\Sigma}^{-1} \hat{\mu}_j + \ln(\hat{\pi}_j)$. Classification chooses the largest δ_j .

Question

For a 2x2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the inverse is:

- A. $\frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$
- B. $\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$
- C. $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$
- D. $\frac{1}{2} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

Answer: A

Explanation

Inverse of 2x2 = $1/\text{determinant} \times \text{adjugate}$. Determinant = $ad - bc$.

Question

The covariance between two features x_1 and x_2 is:

- A. $\frac{\sum (x_1 - \bar{x}_1)(x_2 - \bar{x}_2)}{N - 1}$
- B. $\sum x_1 x_2$
- C. $\bar{x}_1 \cdot \bar{x}_2$
- D. $\frac{\sum x_1}{N} \cdot \frac{\sum x_2}{N}$

Answer: A

Explanation

Covariance = average product of deviations from means, using $N - 1$ for sample covariance.

Question

The Akaike Information Criterion is defined as:

- A. $AIC = 2k - 2 \ln(L)$
- B. $AIC = k + L$
- C. $AIC = 2k + 2 \ln(L)$
- D. $AIC = L/k$

Answer: A

Explanation

$AIC = 2k - 2 \ln(L)$ where k = number of parameters and L = likelihood. Lower AIC = better model.

Question

The Bayesian Information Criterion is defined as:

- A. $BIC = k \ln(N) - 2 \ln(L)$
- B. $BIC = 2k - 2 \ln(L)$
- C. $BIC = k + L$
- D. $BIC = L^2$

Answer: A

Explanation

$BIC = k \ln(N) - 2 \ln(L)$. It penalises complexity more heavily than AIC for large samples.

Question

Heteroskedasticity in regression refers to:

- A. Non-constant variance of residuals across observations
- B. Non-normal residuals
- C. Nonlinearity in relationships
- D. Multicollinearity

Answer: A

Explanation

Heteroskedasticity = variance of errors is not constant. It inflates standard errors and invalidates inference.

Question

Cook's distance is used to:

- A. Measure the influence of each observation on fitted regression coefficients
- B. Measure model accuracy
- C. Detect multicollinearity
- D. Estimate variance

Answer: A

Explanation

Cook's distance measures how much coefficients change if an observation is removed. High values identify influential points.

Question

White's Test in regression is used to:

- A. Test for heteroskedasticity
- B. Test for normality
- C. Test for autocorrelation
- D. Test for linearity

Answer: A

Explanation

White's Test detects heteroskedasticity (non-constant error variance). A significant result indicates heteroskedasticity.